Thermal Shielding, Melt Kinetics and Surge Control of a Steam-Assisted Pipeline Cleaning Crawler: A Monte Carlo Study and Its Deployment Limits

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Highlights

- Probabilistic model of a steam-assisted pipeline crawler: heat shield, melt and surge
- An 18 μm retained melt film already cuts the median range below 100 m
- No single 45 min deployment cleans 300 m; a 1 km line needs about 8 runs
- Heating the pipe wall adds 1.8 times the steam needed to melt the deposit
- A bypass valve closed at the stall cannot arrest the surge impulse

Abstract

Steam-assisted tools could remove paraffin deposits from hydrocarbon pipelines without abrasives, but must shield their electronics from steam near 200 °C, melt the deposit fast enough to be useful, and control speed when a blockage releases. We evaluate a motorized steam-assisted pipeline crawler with a probabilistic reduced-order model: transient conduction through an aluminium–aerogel–titanium wall coupled to a lumped battery pack, film condensation on the deposit, a blockage-release transient with an orifice bypass, and a steam budget. Inputs are assumed design-stage distributions, and every added mechanism can be switched off to recover

the original model exactly. With 40 mm of aerogel the pack stays at or below 55 °C for 45 min in 98.9% of 3,000 draws, but only in a cool pipe: the probability falls to 84% at 35 °C and 22% at 45 °C, and the hull does not fit pipes below 8 in. The cleaned range per deployment, a median of 138 m when molten deposit is removed at once, falls to 27 m with a retained melt film 0.2 mm thick and to 2 m with none removed. No 45 min deployment reaches 300 m, so a 1 km line needs a median of 8 deployments and about 31 h under assumed logistics. Heating the pipe wall adds 1.8 times the melting steam. A bypass that is closed at the stall cannot arrest the surge, because the tool reaches 1 m/s in about 6 ms. The results rank the measurements needed; they are not validated performance.

Keywords

Pipeline pigging; Paraffin deposition; Film condensation; Silica aerogel; Transient heat conduction; Monte Carlo simulation

1. Introduction

Wax, asphaltene, scale and sediment deposits narrow the bore of upstream hydrocarbon pipelines, raise the pressure loss and degrade the product. Wax deposition has been studied extensively [1,2], and the usual remedies are high-volume flushing, chemical treatment and mechanical pigging. Each has a known cost: abrasive or mechanical cleaning can wear the wall and generates solid waste, and a passive pig can stall behind a heavy deposit and then surge, because the gas stored behind it expands when the blockage lets go [3]; bypass valves that vent the gas are used to control the speed of pigs in gas lines [4,5]. Ice-slurry pigging has been developed for water mains and process plant [6,7]. A scouting challenge published by an energy company has recently asked for cleaning solutions for pipelines of 4 to 16 inches and up to 1 km

that have no dedicated launcher or receiver and can be reached only through a connection aligned with the flow [8].

Thermal mobilization with steam is an obvious alternative to abrasion, because condensation delivers a large heat flux at a modest temperature difference. It raises three engineering questions that this paper tests for one concept, a tracked, self-propelled crawler with an elastomeric sealing skirt that melts deposit with steam ahead of it (the Thermally Assisted Motorized Pipeline Remediation Crawler, TAMPRC). First, can the battery and electronics of a tool that sits in saturated steam near 200 °C be kept below a safe temperature for a useful mission? Second, how fast can the deposit be melted, and therefore how far can the tool clean in one deployment? Third, can the on-board drive control the surge when a blockage releases, or is a pressure-relief path needed? A fourth question comes from the published brief: does the concept fit the bore, the length, the ambient temperature and the access that the brief states?

The paper evaluates these questions with a Monte Carlo model, organized as a single seeded script in the spirit of the MyUncle maintenance-lattice framework [9] and propagating assumed input distributions [10]. The first suite (thermal shield, melt kinetics, surge, maintenance lattice) was written as a design-screening study and is retained unchanged in the repository. This paper re-examines its assumptions against the brief and adds five mechanisms that can each be switched off: a retained melt film that resists heat flow, a pipe-length and logistics model, a diameter and ambient-temperature study, an orifice model of the bypass relief that vents the upstream gas, and a steam budget that includes heating the pipe wall. With all of them off the extended model reproduces the first suite draw for draw (Section 5). The contribution that can be defended is a set of quantified limits of the concept, and a list of the measurements that would

remove them, not a predicted performance: all inputs are assumed, and no measurement of these deposits, steam conditions or tool was available.

2. Concept and operating envelope

The tool (Fig. 1) is a crawler of about 120 kg with cleated tracks and an elastomeric sealing skirt. Steam condenses on the deposit face ahead of the skirt, delivers latent heat and melts the deposit in place, and the skirt sweeps the liquefied residue forward. The battery pack, motors and electronics sit in a sealed vessel inside a concentric stack of aluminium shell, silica aerogel [11] and titanium hull; an evacuated (vacuum-jacketed) variant is also evaluated. Brushless motors drive the tracks and act as regenerative brakes against a speed set-point of 0.10 m/s, and a seal-differential bypass valve is added to relieve the pressure when a blockage releases.

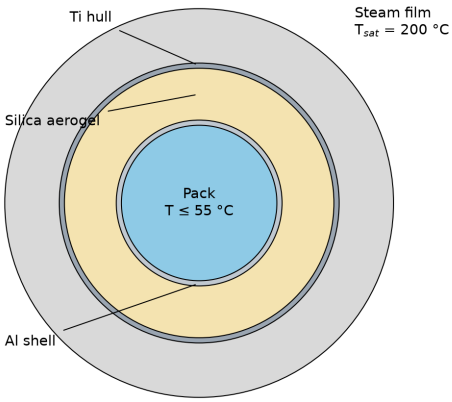


Fig. 1. Cross-section of the crawler hull (not to scale): pack, aluminium shell, silica aerogel, titanium hull and the steam film.

Table 1 lists the operating envelope that the brief states [8] next to what the modelled tool assumes. The brief asks for evidence of removal, compatibility with the connection and recovery arrangements, safety, limited disruption and a proof-of-concept plan, and it excludes purely conceptual proposals and solutions that depend on permanent pigging facilities. The tool

modelled here is at concept stage (technology readiness level 2 to 3), below the 4 to 6 that the brief requires, and this paper does not claim otherwise.

Item	Brief [8]	Modelled tool
Pipe diameter	4 to 16 in	12 in (0.30 m bore) baseline; 4 to 16 in studied in Section 6.5
Pipe length	about 0.3 m to 1 km	one 45 min deployment; 0.3 m to 1 km studied in Section 6.4
Material	carbon steel	carbon steel, standard wall
Deposits	paraffin, sediment, scale, sludge, residue	meltable paraffin only (melt point 58 ± 8 °C); solids not modelled
Access	connection aligned with the flow; no fixed launcher or receiver	not modelled; tool recovery and steam path are open design items
Stages	preparation, isolation, cleaning, residue collection, verification, return to service	active cleaning and assumed logistics durations (Section 6.4)
Readiness	technology readiness level 4 to 6	level 2 to 3 (simulation only)

Table 1. Operating envelope in the published brief and in the model.

Two limitations of the concept are stated at the outset because they decide the interpretation of the results. First, the steam must reach a tool that is sealed against the deposit it is melting; this needs a defined forward path (a lance or bypass annulus) or a steam line behind the tool, and the latter heats the whole pipe wall (Section 6.7). Second, the electronics of a tool in a hydrocarbon line need equipment protection suitable for explosive atmospheres [12], and the saturation pressure at 198 °C, 1.55 MPa, exceeds the pressure rating of a Class 150 carbon-steel flange at 200 °C, which is of the order of 14 bar [13]; neither point is modelled.

3. Mathematical model

3.1 Transient conduction in the composite wall

The radial temperature $T(r,t)$ in the composite wall (aluminium shell, silica aerogel, titanium hull) obeys one-dimensional cylindrical conduction,

$$\rho\ c_p(T)\ \frac{\partial T}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left[r\ k(T)\ \frac{\partial T}{\partial r} \right] \tag{1}$$

with density ρ, specific heat c_p and conductivity k. For atmospheric-pressure aerogel $k=k_0[1+\alpha(T-25)]$ with $k_0=0.018$ W/m·K and a slope of 8×10^{-5} W/m·K^2; for the evacuated jacket

k=0.0040+2.5×10^{-5}(T−25) W/m·K. Both are multiplied by a lognormal factor for compaction, moisture and lot variation. The outer boundary exchanges heat with saturated steam through a film coefficient, the inner boundary with a lumped battery pack (heat capacity $m_{\mathrm{pack}}c_{\mathrm{p}}/L$ per unit length, source P_{gen}/L), and a parallel bridge conductance (tracks, end caps) equal to a fraction $\varphi \sim U(0.05,0.30)$ of the nominal aerogel conductance joins hull and pack. The pack limit is T_{crit}=55 °C; this is an assumed design limit, conservative relative to the onset of self-heating reported for lithium-ion cells [14], and no cell-level analysis is made. Eq. (1) is discretized by finite volumes (2 shell, 16 aerogel and 2 hull cells) and integrated with backward Euler at 10 s, with the conductivity lagged one step.

3.2 Film condensation

The condensation coefficient is the Nusselt film correlation [15,16], adapted to a conduit of diameter D,

$$h = 0.728\left[\frac{g\ \rho_l(\rho_l-\rho_v)\ k_l^3\ h'_{fg}}{\mu_l\ D\ (T_{sat}-T_w)}\right]^{1/4}, \qquad h'_{fg} = h_{fg} + 0.68\ c_{p,l}(T_{sat}-T_w) \tag{2}$$

with saturated-water properties near 1.55 MPa. It was derived for outside horizontal tubes, so its use in a conduit is an assumption, covered by a lognormal multiplier (σ=0.15) and a non-condensable-gas factor $U(0.5,1)$. It scales as $D^{-1/4}$, which the tests verify.

3.3 Deposit melting, and the retained melt film

A deposit of thickness δ and melt point T_{m} needs the energy per unit area $e=\rho_{\mathrm{d}}\delta[L_{\mathrm{f}}+c_{\mathrm{d}}(T_{\mathrm{m}}-T_0)]$. The first suite assumed that molten deposit leaves the surface at once (ablation), so that the heat flux is that of the condensate film at the melt temperature, $q=h(T_{\mathrm{sat}}-T_{\mathrm{m}})$. A liquid layer that stays on the surface adds a conduction resistance in series. With a retained layer of thickness s and liquid conductivity k_{l}, and a fraction $\eta \sim U(0.15,0.50)$ of the flux reaching the deposit,

$$q_{eff}=\eta\ \frac{T_{sat}-T_m}{1/h+s/k_l},\qquad t_m=\frac{e}{q_{eff}},\qquad v_{th}=\frac{L_z}{t_m},\qquad v=\min\left(v_{set},v_{th}\right) \tag{3}$$

where $L_z\sim U(0.3,0.8)$ m is the steam contact length and v_{set}=0.10 m/s. Setting s=0 recovers the first suite exactly. The opposite limit, in which no melt drains and the layer grows from 0 to δ, gives

$$t_{nd}=\frac{\rho_d\left[L_f+c_d\left(T_m-T_0\right)\right]\ \left(\delta/h+\delta^2/2k_l\right)}{\eta\ \left(T_{sat}-T_m\right)} \tag{4}$$

The liquid conductivity is assumed (k_l=0.15 W/m·K). The cleaned range of a deployment is v min (t_{crit},t_{run}), where t_{crit} is the first time the pack reaches 55 °C and t_{run} the mission length (45 min).

3.4 Blockage release with an orifice bypass

From break-through the tool of mass m obeys

$$m\ \frac{dv}{dt}=A\ \ \Delta P\ \left(1-\beta\right)-F_f-c_d v^2-F_c \tag{5}$$

with pipe area A, seal friction F_f, downstream drag c_d and controller force F_c (PI on speed, limited to the braking and traction authority). The first suite represented the bypass relief by a factor $\beta\rightarrow\beta_{max}$[1−0.5 bar/Δ$P$] (0.85 to 0.97) with a 5 to 30 ms lag, that is, as a force reduction without a flow. Here the upstream gas pocket (initial volume V_0, pressure P_0=P_d+ΔP_b) expands adiabatically as the tool advances a distance x and loses its gas through the bypass,

$$P_u=P_0\left(\frac{\varphi V_0}{V_0+A\ \ x}\right)^{\gamma},\qquad \frac{d\varphi}{dt}=-\frac{\dot{m}_b}{m_0},\qquad \dot{m}_b=C_d A_b\ \ u\ \ P_u\sqrt{\frac{2\gamma}{\left(\gamma-1\right)RT}}\sqrt{r^{2/\gamma}-r^{\left(\gamma+1\right)/\gamma}} \tag{6}$$

where φ is the remaining gas fraction, m_0=P_0V_0/RT, r=max (P_d/P_u,r_c) with r_c the critical ratio (choked flow), A_b the bypass flow area, C_d=0.7 and $u$in[0,1] the valve opening, which follows the demand (ΔP−0.5 bar)/(0.5 bar) with the same 5 to 30 ms lag. The gas is methane-like (R=518 J/kg·K, T=300 K, γ=1.3, assumed) and the downstream pressure is held constant. With A_b=0 the

model reduces to the first suite's crawler without relief, and with φ=1 the pocket pressure is the same function of x.

3.5 Pipe length, logistics and steam budget

For a line of length L and a deployment range $R_{\mathrm{dep}}=v\,t_{\mathrm{run}}$, the number of deployments is the smallest integer not below L/R_{dep}, the active cleaning time is L/v, and the total intervention time adds preparation and isolation, turnaround between deployments, verification and return to service, whose durations are assumed (Table 2). If steam reaches the tool through the pipe, it must also heat the steel wall of the whole line behind it,

$$\dot{m}'_{s}=\frac{\rho_s A_s c_s\,\left(T_{sat}-T_{p}\right)}{h_{fg}}+\dot{m}'_{melt} \tag{7}$$

per metre of line, with the steel cross-section A_{s} of standard-wall pipe (ρ_{s}=7,850 kg/m^3, c_{s}=490 J/kg·K), the pipe temperature T_{p} and the melting steam $\dot{m}'_{\mathrm{melt}}$ of Section 3.3. The radial clearance (tracks, cleats, skirt) is that of the first design in a 12 in pipe, 44 mm; the hull is scaled for other sizes by reducing the pack radius, and then the aerogel, down to a 25 mm pack. The maintenance lattice of the first suite (a periodic 48×24 lattice of wall cells fouled at a rate $r_{\mathrm{f}}(1+\gamma n_{\mathrm{f}}/4)$ and cleared with probability p per pass, checked against its exact solution for γ=0 [9]) is retained and reported in Section 6.8; it is not calibrated to any deposit.

4. Monte Carlo methodology

All inputs are sampled independently from the distributions in Table 2 with fixed seeds. The thermal studies use 3,000 draws per scenario (800 for the sweeps), the surge study 10,000, and probabilities carry Wilson 95% intervals [17]. Sensitivity is reported as the Spearman rank correlation of each input with the output [18]. Common random numbers are used across the retained-film, coverage and steam studies, so differences between cases are not sampling noise.

The distributions are assumed design-stage values, not measurements; no published distribution of these quantities for the deposits, steam conditions or tool was found.

Input	Distribution or value
Aerogel thickness; k multiplier	40 mm × (1 + N(0, 0.03)) (swept 5–60 mm); lognormal σ = 0.15 (× lognormal σ = 0.30 and 5% vacuum loss when evacuated)
Pack mass; heat generation; initial temperature	N(16, 1.2) kg; lognormal, median 70 W, σ = 0.35; N(25, 4) °C (shifted 0–55 °C in Section 6.5)
Thermal-bridge fraction; condensation multiplier; non-condensable factor	U(0.05, 0.30); lognormal σ = 0.15; U(0.5, 1.0)
Steam saturation temperature	N(198, 3) °C truncated to 185–204 °C
Deposit thickness; melt point; density; latent heat	lognormal, median 6 mm, σ = 0.5 (1–40 mm); N(58, 8) °C; N(850, 40) kg/m³; N(200, 30) kJ/kg
Pipe temperature; heat-coupling fraction η; contact length	N(25, 5) °C; U(0.15, 0.50); U(0.3, 0.8) m
Retained melt film s; liquid conductivity	swept 0–5 mm (not sampled); 0.15 W/m·K
Break-through ΔP; gas-pocket volume; downstream pressure	lognormal, median 3 bar, σ = 0.6 (0.5–25 bar); lognormal, median 1 m³, σ = 0.7 (0.2–10 m³); U(5, 30) bar
Seal friction; downstream drag; masses	lognormal, median 400 N; lognormal, median 2,000 N s²/m²; N(80, 8) kg pig, N(120, 10) kg crawler
Braking limit; traction limit; relief lag	N(2.5, 0.4) kN; N(1.5, 0.2) kN; U(5, 30) ms
Bypass area; discharge coefficient; crack differential	swept 0–320 cm² (pipe 707 cm²); 0.7; 0.5 bar
Assumed logistics durations (h)	preparation and isolation U(3, 8); turnaround between deployments U(1, 3); verification U(1, 3); return to service U(2, 6)

Table 2. Principal sampled or assumed inputs.

5. Verification

The model is verified numerically; it is not validated physically, because no measurement exists for these inputs. Twenty-seven automated checks run on the committed code and results (tests/test_all.py). They confirm that the first suite is intact (the file is byte-identical to the one submitted, and rerunning it reproduces its published results to the last digit); that every added mechanism reduces to the first suite when switched off (the extended thermal, melt and surge functions equal the originals draw for draw, and the orifice model with zero area equals the controller-only crawler); and limiting and conservation properties of the added mechanisms: the series-resistance flux, the $D^{-1/4}$ scaling of the condensation coefficient, zero flow at zero differential and choking below the critical ratio, a gas fraction that stays between 0 and 1,

venting times that scale with pocket volume, and the energy balance of the wall-heating steam. A deterministic nominal case refined from 20 to 160 cells and from 10 s to 1.25 s changes the pack temperature at 45 min from 38.65 °C to 38.68 °C, a difference of 0.03 K, and the differences to the finest grid fall monotonically (0.025, 0.010 and 0.003 K). The exact mean-field solution of the maintenance lattice agrees with the lattice Monte Carlo within 0.26 percentage points for γ=0.

6. Results

6.1 Thermal shield

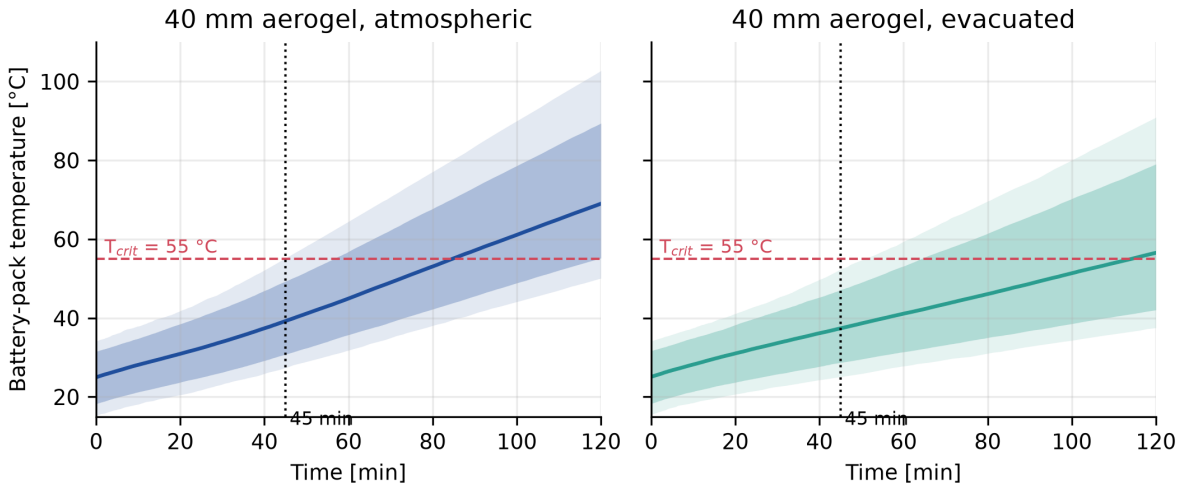


Fig. 2. Pack temperature against time for 40 mm of aerogel, atmospheric (left) and evacuated (right): median, 5–95% and 1–99% bands.

At 45 min the pack median is 39.2 °C (95th percentile 49.3 °C) with atmospheric aerogel and 37.4 °C (47.1 °C) with the evacuated jacket (Fig. 2). The probability of staying at or below 55 °C is 98.9% (95% interval 98.4–99.2%) and 99.7% (99.5–99.9%) respectively. The median time to reach 55 °C is 85 min (5th percentile 56 min) and 114 min (64 min), and 44% of the evacuated draws never reached it within the 120 min horizon. The first suite's 3,000 draws and the re-drawn 3,000 of the extended suite agree (98.8% against 98.9%).

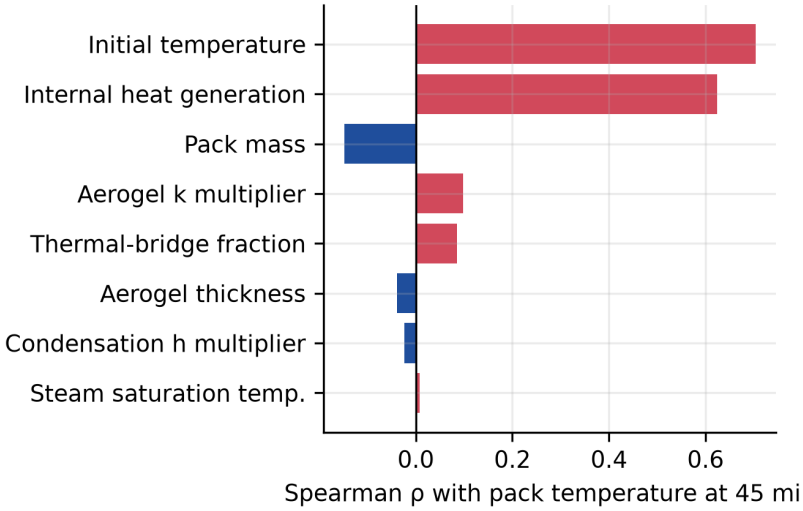


Fig. 3. Sensitivity of the 45 min pack temperature (atmospheric case): Spearman rank correlation with each input.

Initial temperature ($\rho = 0.70$) and internal heat generation (0.62) dominate; pack mass is next (−0.15), and aerogel conductivity (0.10), bridge fraction (0.09) and thickness (−0.04) are weaker at the 40 mm design point, because the diffusion time of aerogel across 40 mm is of the order of hours, so the wall has not reached steady state within the mission (Fig. 3). The thermal margin therefore belongs to the thermal mass of the pack and to the starting temperature, not to the insulation alone.

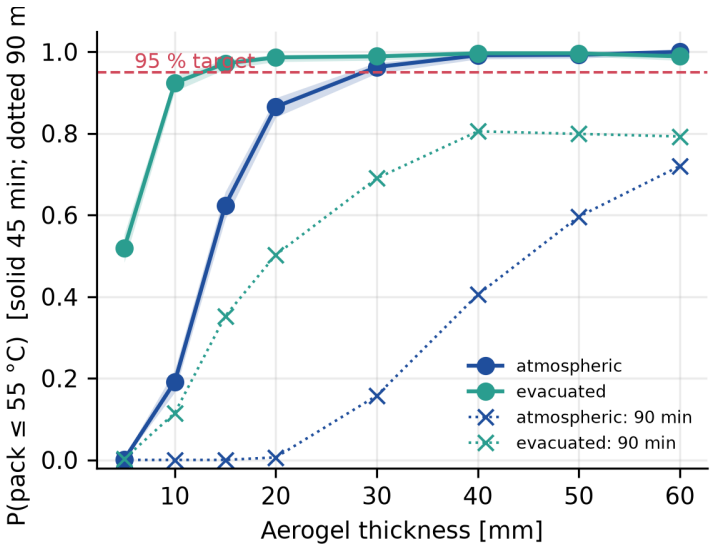


Fig. 4. Probability of meeting the 55 °C limit against aerogel thickness (solid: 45 min; dotted: 90 min).

A 95% reliability target at 45 min requires roughly 30 mm of atmospheric aerogel (96.2%) or 15 mm of evacuated aerogel (97.1%) (Fig. 4). The evacuated curve plateaus near 99% because of the assumed 5% vacuum-loss probability and the tails of initial temperature and heat generation. For a 90 min mission the probability at 40 mm falls to 40% (atmospheric) and 80% (evacuated).

6.2 Melt kinetics and range, with instant removal of the melt

With the first suite's assumption that molten deposit leaves the surface at once, the median condensation coefficient is 3.0 kW/m²·K and the film flux 413 kW/m². The melt time of a deposit has a median of 10.8 s (95th percentile 32 s). The thermally limited speed has a median of 5.1 cm/s (5th percentile 1.5 cm/s) and is below the 10 cm/s set-point in 83% of the draws. Deposit thickness is its strongest driver ($\rho = -0.68$), followed by the heat-coupling fraction (0.46) and the contact length (0.37). The median cleaned range per 45 min deployment is 138 m (5th to 95th percentile 39 to 270 m; the upper value is the set-point cap of 0.1 m/s × 2,700 s). The probability of a range of at least 100, 200 and 250 m is 66%, 30% and 20%. The median

steam use is 1.75 kg per metre (95th percentile 4.8 kg/m), about 0.18 m³ of condensate per 100 m (Fig. 5).

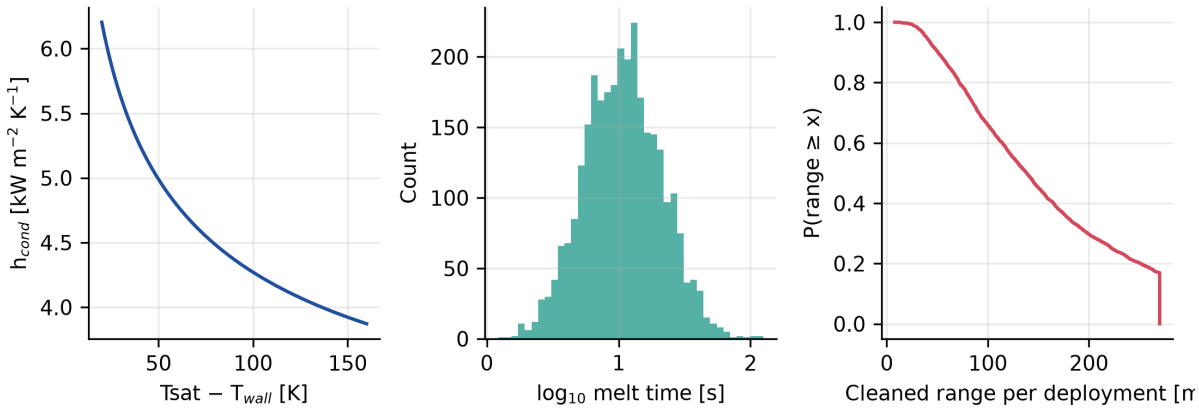


Fig. 5. Condensation coefficient against temperature difference, melt-time distribution and exceedance of the cleaned range per deployment (instant removal of the melt).

6.3 The retained melt film decides the range

The first suite's melt model is the optimistic bound of a family. Molten paraffin that stays on the surface adds the resistance s/k_l to the condensate film $1/h$ (Eq. 3), and since $k_l/h = 0.051$ mm, a retained layer thicker than about 51 µm already halves the heat flux. Table 3 and Fig. 6 give the range against the retained thickness for the same draws. The median range falls to 68 m at 0.05 mm, 27 m at 0.2 mm and 6.6 m at 1 mm; the median range drops below 100 m at a retained film of only 18 µm. If none of the melt drains (Eq. 4), the median melt time is 629 s instead of 10.6 s, a factor 59, and the median range is 2.3 m. The heat-coupling fraction η is applied on top of the film, so these ranges include the first suite's unexplained heat loss as well; they are not a calibrated prediction. They show that whether the melt leaves the surface is the single most influential unmeasured assumption of the concept: the reported 138 m is its upper end and not an expected value.

s (mm)	Median v_th (cm/s)	Median range (m)	5th to 95th percentile (m)	P(range ≥ 100 m)	Thermally limited
0 (ablation)	5.0	134	40 to 270	65%	83%
0.05	2.5	67.8	21 to 213	29%	98%
0.10	1.7	45.5	14 to 140	12%	100%
0.20	1.0	27.4	8.4 to 83	3%	100%
0.50	0.46	12.5	3.8 to 38	0%	100%
1.00	0.24	6.6	2.0 to 20	0%	100%
2.00	0.12	3.4	1.0 to 10	0%	100%
5.00	0.05	1.4	0.4 to 4	0%	100%
no drainage (limit)	0.08	2.3	–	0%	100%

Table 3. Median speed and cleaned range per 45 min deployment against the retained melt film s (atmospheric aerogel, 40 mm; same draws in every row).

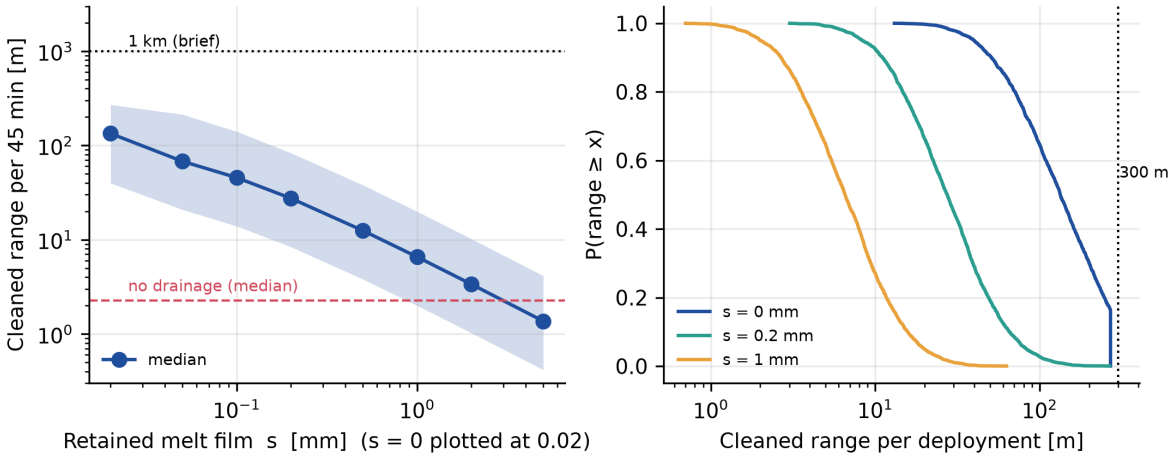


Fig. 6. (a) Cleaned range per 45 min deployment against the retained melt film (median and 5–95% band; dashed: no-drainage limit; dotted: 1 km). (b) Exceedance of the range for three film thicknesses.

6.4 Pipe length, deployments and intervention time

The brief names pipes from about 0.3 m to 1 km. Under the assumed set-point of 0.10 m/s, a 45 min deployment cannot exceed 270 m in any draw, so no draw cleans a 300 m line in one deployment, even with instant removal of the melt (Table 4, Fig. 7). A 1 km line needs a median of 8 deployments (95th percentile 26) with instant removal, 81 with a 0.5 mm retained film and 442 if no melt drains. A 90 min evacuated mission raises the cap to 540 m and the one-

deployment probability for 300 m to 41%, but still does not reach 1 km. With the assumed logistics durations (Table 2) the total intervention time for 1 km has a median of 31 h, of which only 5.6 h (18%) is active cleaning; for 100 m it is 13 h. Because the logistics durations are assumed, the absolute hours matter less than the ratio: most of the intervention is not cleaning. Moreover, every extra deployment requires that the tool is recovered through the same flow-aligned connection by which it entered, which the brief states as the only access; the model does not include recovery or the passage of the tool through bends, valves and diameter changes.

Mission	Melt removal	L (m)	P(one deployment)	Deployments	Active (h)	Total (h)	Active share
45 min, atmospheric	instant removal	100	65%	1 (3)	0.6	13	5%
45 min, atmospheric	instant removal	300	0%	3 (8)	1.7	17	10%
45 min, atmospheric	instant removal	1,000	0%	8 (26)	5.6	31	18%
45 min, atmospheric	0.5 mm film	100	0%	9 (27)	6.0	32	18%
45 min, atmospheric	0.5 mm film	300	0%	25 (79)	18.0	75	23%
45 min, atmospheric	0.5 mm film	1,000	0%	81 (263)	60.0	227	26%
90 min, evacuated	instant removal	100	89%	1 (2)	0.6	12	5%
90 min, evacuated	instant removal	300	41%	2 (5)	1.7	15	12%
90 min, evacuated	instant removal	1,000	0%	4 (14)	5.6	24	23%
90 min, evacuated	0.5 mm film	100	2%	5 (14)	6.0	25	24%
90 min, evacuated	0.5 mm film	300	0%	13 (42)	18.1	53	33%
90 min, evacuated	0.5 mm film	1,000	0%	43 (140)	60.2	153	38%

Table 4. Deployments and time to clean a line of length L (median; 95th percentile of the deployment count in parentheses). Durations of the non-cleaning stages are assumed (Table 2).

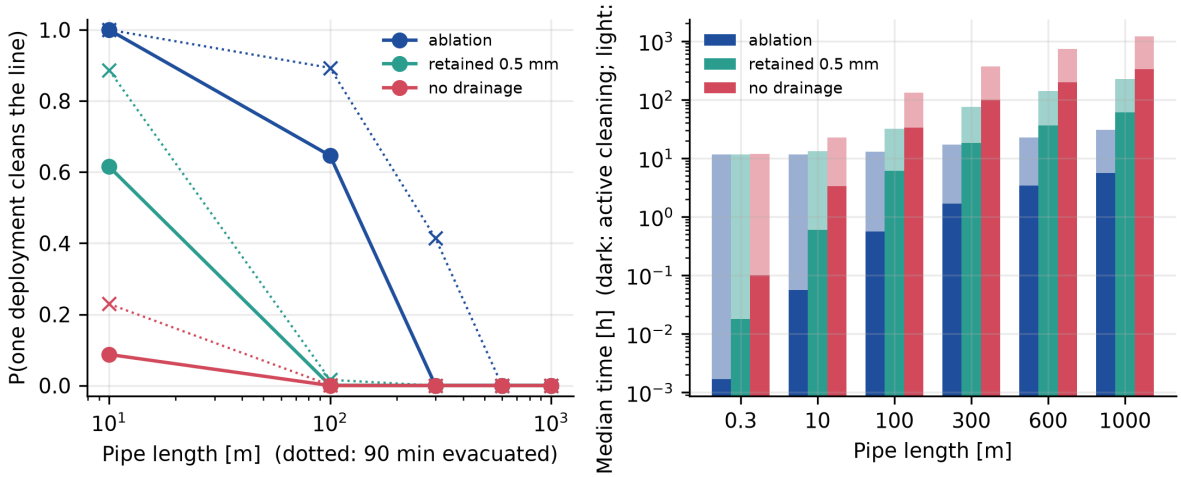


Fig. 7. (a) Probability that one deployment cleans a line against its length for three melt-removal cases (solid: 45 min atmospheric; dotted: 90 min evacuated). (b) Median active cleaning time (dark) and total intervention time (light), 45 min atmospheric.

6.5 Pipe size and pipe temperature

The brief covers 4 to 16 inches; the first design was sized for 12 in. Keeping the radial clearance of the first design (44 mm), the hull does not fit in a 4 or 6 in pipe, even with the aerogel removed down to a 25 mm pack. In an 8 in pipe the aerogel shrinks to 24 mm and the pack to 2.8 kg, and the probability of keeping the pack at or below 55 °C for 45 min is 0% (5% evacuated); in a 10 in pipe it is 42% (50% evacuated); from 12 in upward it is 99% (Table 5, Fig. 8). The thermal margin therefore scales with the thermal mass of the pack, and the architecture as modelled covers about the upper third of the brief's diameter range. The surge force of a 3 bar break-through grows with the bore, from 10 kN (8 in) to 35 kN (16 in), against the same 2.5 kN of braking authority.

NPS (in)	Bore (mm)	Hull OD fitted (mm)	Aerogel (mm)	Pack (kg)	P(≤ 55 °C) atm.	P(≤ 55 °C) evac.	Force at 3 bar (kN)	Force / braking
4	102	–	does not fit	–	–	–	2.5	1.0
6	154	65	does not fit	–	–	–	5.6	2.2
8	203	114	24	2.8	0%	5%	9.7	3.9
10	254	166	40	5.4	42%	50%	15.3	6.1
12	305	216	40	16.0	99%	99%	21.9	8.8

NPS (in)	Bore (mm)	Hull OD fitted (mm)	Aerogel (mm)	Pack (kg)	P(≤ 55 °C) atm.	P(≤ 55 °C) evac.	Force at 3 bar (kN)	Force / braking
16	387	298	40	16.0	99%	99%	35.4	14.1

Table 5. Fit and thermal margin against nominal pipe size (standard wall; 45 min; 40 mm aerogel where it fits).

The pack starts at the pipe temperature, which is the dominant input (Section 6.1). Shifting the initial pack and pipe temperature with the same draws, the probability for the 12 in design falls from 99% at 25 °C to 84% at 35 °C, 22% at 45 °C and 0% at 55 °C (atmospheric; evacuated 100%, 90%, 36% and 0%), while the median range changes little (133 m at 25 °C, 102 m at 45 °C) until the pack limit is reached early. A line that still carries warm produced fluid, for example at 40 to 60 °C, must therefore be cooled and flushed before the tool enters, which is a preparation step that the intervention time of Table 4 only partly includes.

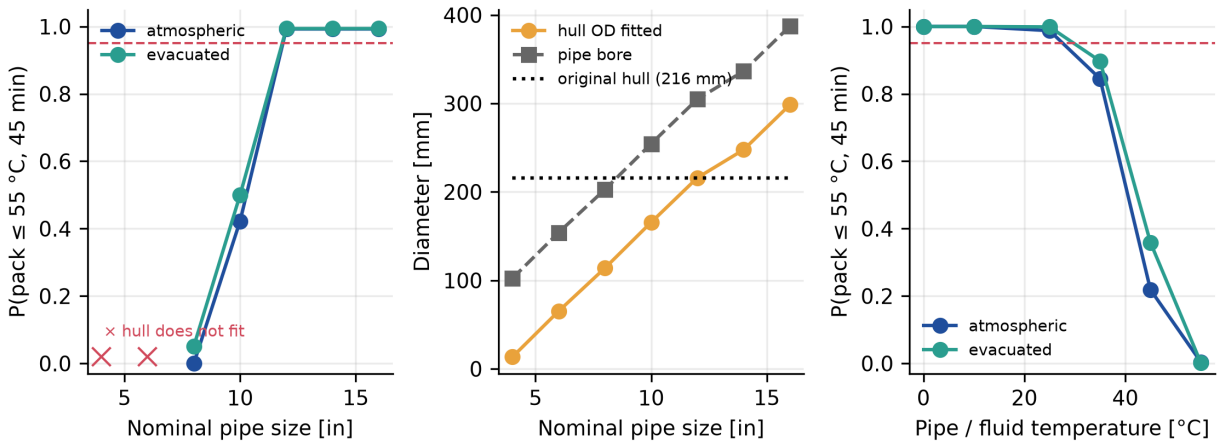


Fig. 8. (a) Probability of keeping the pack at or below 55 °C for 45 min against nominal pipe size (crosses: the hull does not fit). (b) Fitted hull diameter, pipe bore and the first design's hull. (c) The same probability against pipe temperature, 12 in design.

6.6 Surge on blockage release

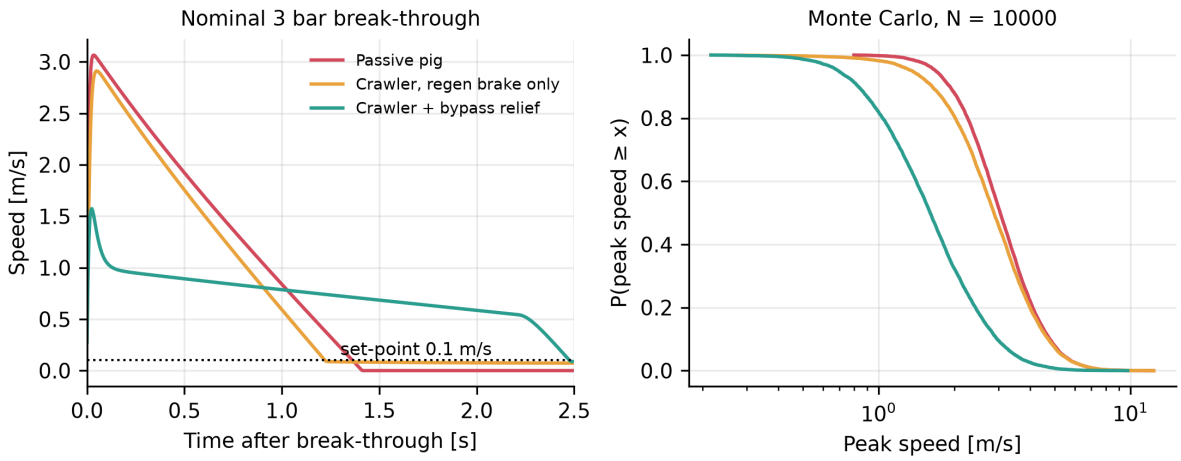


Fig. 9. Speed transient for a nominal 3 bar break-through (left) and Monte Carlo exceedance of the peak speed (right), first suite.

In the first suite the pressure force of a 3 bar break-through on a 12 in bore, $A\,\Delta P \approx 21$ kN, exceeds the 2.5 kN braking authority by about a factor of nine, so braking alone changes the median peak speed only from 3.0 to 2.9 m/s (Table 6, Fig. 9). Treating the bypass relief as a force factor β, the median peak falls to 1.6 m/s. Venting the gas pocket through an orifice (Eq. 6) gives a smaller benefit: with 80 cm² of bypass area, 11% of the bore, the median peak is 2.5 m/s, and even with 320 cm², 45% of the bore, it is 2.1 m/s with 53% of the releases above 2 m/s, against 32% in the force-factor model. The reason is a mismatch of time scales. The tool reaches 1 m/s in 6 ms (5th percentile 2 ms), whereas a fully open bypass of 80 cm² needs 64 ms to halve the differential across a 1 m³ pocket (19 ms for 0.3 m³, 635 ms for 10 m³), and a 20 ms valve lag is added to that. The relief therefore limits the duration of the surge, bringing the speed back to the set-point within about 0.3 s for 80 cm² (Fig. 10b), but not its first peak. A bypass that is closed until the differential reaches the break-through value cannot arrest the impulse. The design alternative is a bypass that is already open during the stall, so that the pocket never

charges: it holds a differential of 0.5 bar at 18 bar upstream if its area passes the supply flow, which requires 5 cm² for 0.03 m³/s, 16 cm² for 0.1 m³/s and 47 cm² for 0.3 m³/s (assumed methane-like gas; Fig. 10a). The price is that the differential that melts or breaks the deposit is then limited as well, so the deposit must be freed thermally rather than by pressure.

Configuration	Median peak (m/s)	P(> 1 m/s)	P(> 2 m/s)	Model
Passive pig	3.04	100%	88%	first suite
Crawler, braking only	2.87	98%	81%	first suite
Crawler + relief as force factor β	1.60	82%	32%	first suite
Crawler + orifice bypass 10 cm²	2.80	98%	78%	this paper
Crawler + orifice bypass 20 cm²	2.75	97%	77%	this paper
Crawler + orifice bypass 40 cm²	2.67	96%	73%	this paper
Crawler + orifice bypass 80 cm²	2.54	94%	69%	this paper
Crawler + orifice bypass 160 cm²	2.34	91%	62%	this paper
Crawler + orifice bypass 320 cm²	2.08	86%	53%	this paper

Table 6. Peak speed after break-through (12 in bore; same draws; N = 10,000).

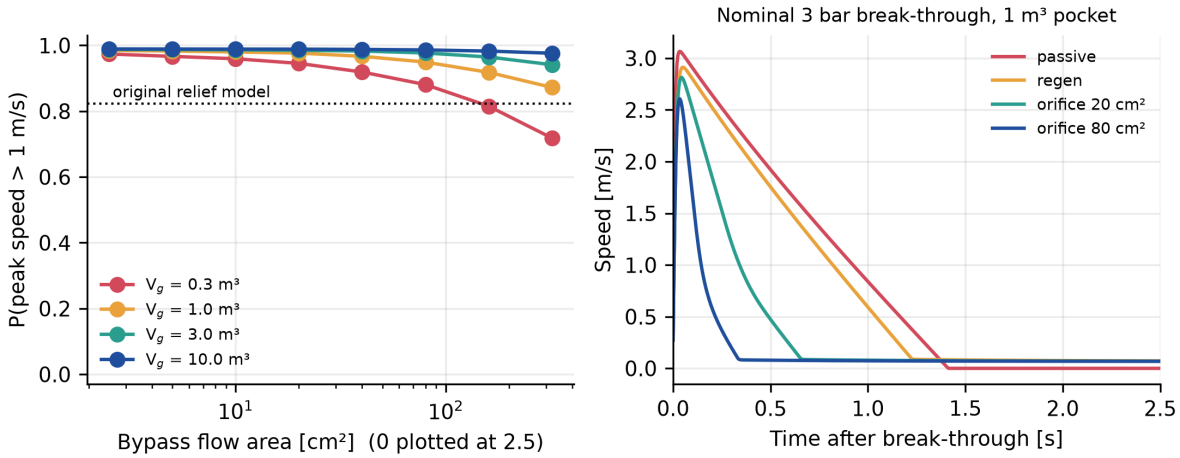


Fig. 10. (a) Probability of a peak speed above 1 m/s against the bypass area for four gas-pocket volumes (dotted: the first suite's force-factor model). (b) Speed after a nominal 3 bar break-through with a 1 m³ pocket.

6.7 Steam budget

Melting the deposit uses a median of 1.79 kg of steam per metre in a 12 in pipe (first suite). If the steam reaches the tool through the pipe behind it, the steel wall of the whole line must also be

heated from 25 °C to the saturation temperature: for standard-wall 12 in pipe of 74 kg/m this takes 3.2 kg of steam per metre, 1.8 times the melting steam, and the total is 5.0 kg/m, or about 5.0 t of steam (and of condensate) for a 1 km line, against the 1.8 t that the melting alone would suggest (Table 7, Fig. 11). A pipe at 0 °C needs 3.7 kg/m for the wall. The wall-heating term is a lower bound, because it ignores heat lost from the pipe to its surroundings during the intervention. A steam hose carried inside the pipe avoids heating the wall but adds tether drag over up to 1 km and a second 200 °C line to the tool. Either way the water use is larger than the melting alone suggests, which matters because reduced water use is one of the brief's nice-to-have criteria.

NPS (in)	Steel (kg/m)	Melting (kg/m)	Wall heating (kg/m)	Total (kg/m)	Wall heating from 0 °C (kg/m)
4	16	0.64	0.70	1.34	0.80
6	28	0.93	1.23	2.17	1.41
8	43	1.21	1.86	3.07	2.13
10	60	1.51	2.63	4.14	3.01
12	74	1.79	3.23	5.02	3.69
14	81	1.97	3.55	5.52	4.07
16	93	2.25	4.07	6.33	4.66

Table 7. Steam per metre of line: melting (median) and heating the pipe wall (standard wall, 25 °C start; last column 0 °C start).

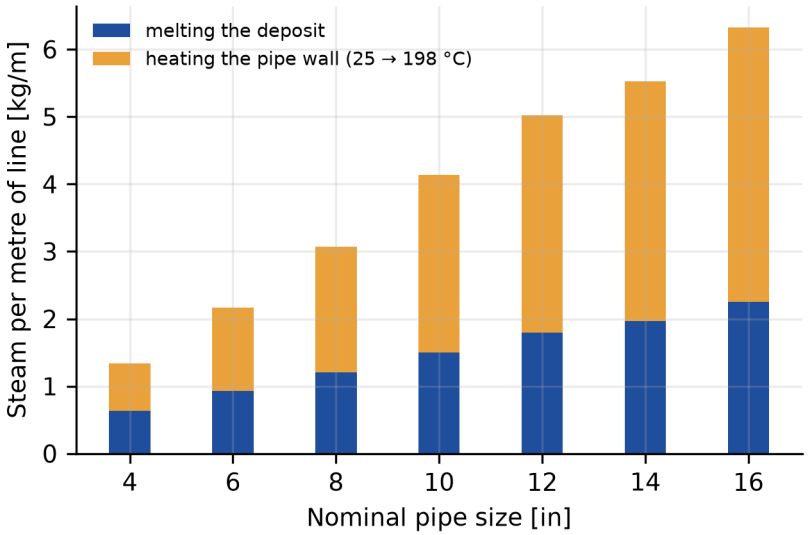


Fig. 11. Steam per metre of line against nominal pipe size: melting the deposit and heating the pipe wall.

6.8 Maintenance lattice

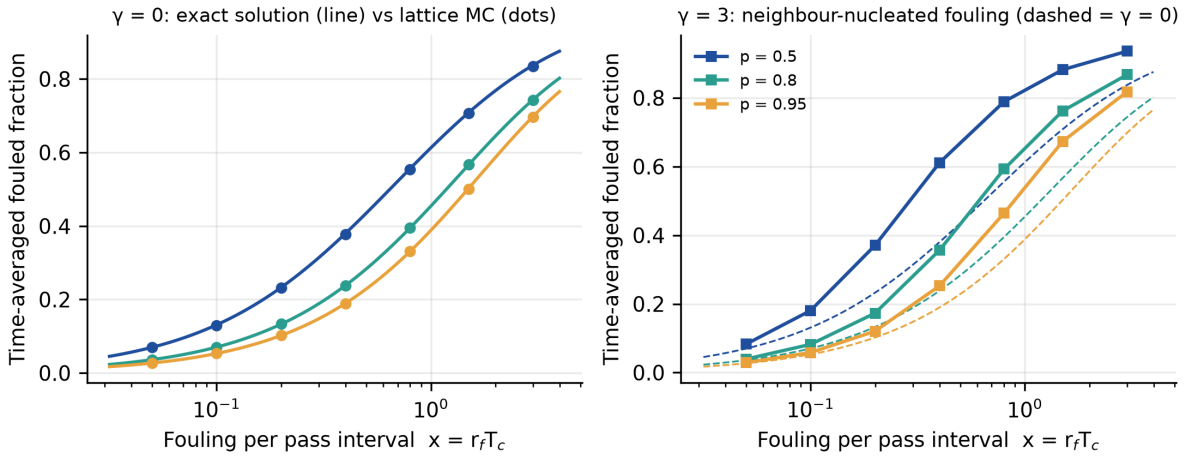


Fig. 12. Time-averaged fouled fraction against fouling per pass interval for three clearing probabilities: non-interacting case (left) and neighbor-nucleated fouling, γ = 3 (right).

For γ=0 the lattice Monte Carlo agrees with the exact solution within 0.26 percentage points. Neighbor-nucleated fouling raises the fouled fraction substantially: for x=0.8 and p=0.8 it rises from 0.40 to 0.59, so a cleaning interval that suffices for independent fouling may fail when

deposits nucleate on existing deposits (Fig. 12). The lattice is not calibrated to any deposit growth rate and is reported as a planning illustration only.

7. Discussion

7.1 What the concept has to demonstrate

The brief states six must-have requirements [8]. Table 8 sets what this study can and cannot say about each. The strongest statement is negative and quantitative: with the assumptions of the first design, the concept does not by itself meet the length, diameter, temperature and access conditions of the brief, and its reach is governed by a quantity, the retained melt film, that nobody has measured. The strongest positive statement is narrower: for a 12 in pipe at 25 °C, 40 mm of aerogel keeps the pack below 55 °C for 45 min in 99% of the draws, so the thermal shield is not the first obstacle, provided the pipe is cool.

Requirement	What the simulation shows	What is missing
Effective removal, with evidence and limits	Range of 138 m (instant melt removal) falling to 27 m (0.2 mm retained film) and 2.3 m (no drainage) per 45 min deployment	Measured melt removal on steel; deposit composition (solid fraction); scale and sediment are not melted
Compatibility with operating conditions, connections and recovery	No single 45 min deployment reaches 300 m; a 1 km line needs a median of 8 deployments; hull does not fit below 8 in	A tool-recovery route through a flow-aligned connection; steam path; bends, valves and diameter changes
Operational safety and integrity	Surge peak not arrested by a bypass that is closed at the stall (tool at 1 m/s in 6 ms); pack margin falls above 35 °C	Explosive-atmosphere protection, steam pressure against the line rating, seal elastomers in hydrocarbon and steam, cell-level battery analysis
Limited disruption, with a full timeline	Active cleaning is 10% to 18% of the intervention (assumed logistics); steam is 5.0 kg/m including the wall	Real preparation, isolation and return-to-service durations
Detailed proof-of-concept plan	Section 7.2 lists the tests, in order of how much each one changes the answer	Costs and schedule are not estimated here
Credibility of the solver	Computational study only; the author is an independent researcher with no hardware or field experience in pipeline cleaning	A partner with pipeline-intervention experience

Table 8. Requirements of the published brief [8] and the evidence of this study.

7.2 Measurements that matter, in order

The Monte Carlo study shows where a measurement changes the conclusion most. First, melt removal: a bench test in which paraffin of known melting point on a steel wall is melted by

condensing steam, with the retained film thickness and the heat flux measured. It decides whether the range is of the order of 100 m or of a few metres (Section 6.3). Second, the thermal soak of a sealed aerogel module with a dissipating pack in saturated steam, at several starting temperatures, which would replace the assumed heat generation and bridge conductance and settle the 55 °C margin (Sections 6.1 and 6.5). Third, a steam-delivery test in a short pipe loop, which measures the condensate and wall-heating mass per metre and shows how steam reaches the tool (Section 6.7). Fourth, a bypass and seal rig for the stall and release, with the valve open before the stall, which measures the differential that can be held and the first-peak speed (Section 6.6). Fifth, characterization of the deposits that the line actually contains, in particular the fraction that does not melt: steam alone does not remove scale or sediment, and the model does not represent the mechanical removal of such solids by the skirt and tracks.

7.3 Limitations

(i) All inputs are assumed, and the retained film thickness, the liquid conductivity, the heat-coupling fraction and the logistics durations are the least constrained. (ii) The conduction model is radial; axial conduction, end caps and track mounts enter through a bridge fraction. (iii) The Nusselt correlation is used outside its derivation range and the steam is taken at rest. (iv) The surge model is reduced-order: a gas pocket vented to a constant downstream pressure, with a quadratic drag standing for the compressibility downstream; liquid-filled lines, which would surge differently, are not modelled. (v) The pack is lumped, with no cell-level thermal runaway analysis. (vi) The steam delivery path, the recovery of the tool, bends, valves and diameter changes are not modelled. (vii) Deposits are meltable paraffin only. (viii) The comparison with ice-slurry pigging [6,7], which the brief names as of particular interest, is not made. The results are a design-screening tool and not a validated prediction.

8. Conclusions

1. The thermal shield is credible but depends on a cool start. With 40 mm of atmospheric aerogel the pack stays at or below 55 °C for 45 min in 98.9% of the draws in a 25 °C pipe, and the probability falls to 84% at 35 °C and 22% at 45 °C; an evacuated jacket helps little at 45 °C (36%).

2. The reach per deployment is governed by whether molten deposit leaves the surface. The median range is 138 m with instant removal, 27 m with a retained film of 0.2 mm and 2.3 m if none drains; a film of 18 μm already brings the median below 100 m. The retained film should be measured before any reach is claimed.

3. Against the brief the concept fails on length, diameter and temperature as modelled: no 45 min deployment reaches 300 m, a 1 km line needs a median of 8 deployments, the hull does not fit pipes below 8 in, and heating the pipe wall makes the steam budget 1.8 times the melting steam.

4. A bypass valve that is closed at the stall cannot arrest the first surge peak, because the tool reaches 1 m/s in about 6 ms while a fully open 80 cm^2 bypass needs about 64 ms to halve the differential over a 1 m^3 gas pocket; a bypass that is open during the stall, sized to the supply flow, is the design alternative.

5. These are the results of assumed inputs. Their value is to rank the measurements: melt removal, a thermal soak in steam, steam delivery and a bypass rig, then deposit characterization.

Declarations

Author contributions (CRediT): Leon Sandler, sole author: conceptualization, methodology, software, formal analysis, writing – original draft, writing – review and editing.

Funding: this work received no external funding.

Competing interests: the author declares no competing interests. The concept was prepared as a response to a public scouting challenge [8]; the author has no financial relationship with its seeker or platform.

Data and code availability: the code, the data of every figure and table and the manuscript are archived at https://doi.org/10.5281/zenodo.24045599 (software, all versions) and https://doi.org/10.5281/zenodo.24045595 (manuscript, all versions), and are available at https://github.com/sandlerleon/tamprc-pipeline-crawler-mc. One command per script regenerates every number and figure; the results quoted in the text are read from the JSON files that the scripts write.

Use of generative AI: a generative AI tool (Claude, Anthropic) assisted with code development, numerical experiments, drafting and editing. The author checked all of it and takes full responsibility for the content.

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